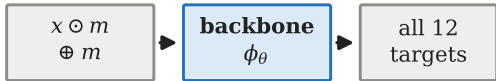


Stage 1: random-mask pre-learning

(once per market)



- $m \sim$ stratified uniform
- $m \sim \text{Bernoulli}(0.5)$
- no masking (ablation)

*masking is necessary;
its distribution is not critical*

Stage 2: subset-specific adaptation

(per query, closed form)



$$Z_{y,S} = [x \odot m_S \mid (x \odot m_S)^2 \mid \phi_1 \mid \dots \mid \phi_R]$$

solved per (S, y) on train only; test used for R^2 alone

contrast: shared output head

one set of output weights for all $2^{|G|}$ masks
 $\Rightarrow 3.2 \times$ / $4.6 \times$ larger V error